RAVADA KARAN

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Linkedin | GitHub | Portfolio

EDUCATION

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science and Engineering — 2022 – 2026

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, PHP, SQL, HTML5, CSS3

Frameworks & Libraries: FastAPI, Flask, Streamlit, OpenCV, YOLOv8, Scikit-Learn, Pandas, NumPy

Databases & Developer Tools: MySQL, PostgreSQL, SQLite, Git, GitHub, RESTful APIs, Postman

Core Competencies: Machine Learning Pipelines, Computer Vision, Full-Stack Web Development, OOP

EXPERIENCE

Navodita Infotech — Remote, India
Full-Stack Web Developer — Dec 2025 – Jan 2026

- Engineered and maintained responsive, full-stack web applications utilizing modern front-end and back-end frameworks.
- Designed and integrated secure RESTful APIs to facilitate seamless data communication with relational database systems.
- Optimized database queries and front-end rendering, significantly reducing API latency and improving page load responsiveness.
- Collaborated in Agile sprints, taking part in code reviews, integration testing, and bug tracking to maintain high software reliability.

PROJECTS

Intelligent Traffic Management System | *Python, YOLOv8, FastAPI, OpenCV, NumPy*

- Constructed a real-time edge AI traffic control system using YOLOv8 for vehicle classification and density tracking.
- Engineered dynamic signal timing algorithms based on real-time vehicle density to mitigate urban intersection congestion.
- Implemented an automated ambulance priority override protocol that clears transit corridors with zero human intervention.
- Developed asynchronous REST APIs using FastAPI for live video stream telemetry processing and traffic monitoring.

Multiple Disease Prediction System | *Python, Scikit-Learn, Streamlit, Pandas*

- Trained and deployed diagnostic ML models (Random Forest, SVM, Logistic Regression) predicting Diabetes, Heart Disease, and Parkinson's.
- Built an intuitive Streamlit web interface integrated with serialized models for sub-second risk factor evaluations.
- Implemented rigorous data preprocessing, feature engineering, and cross-validation pipelines to improve model sensitivity.

Kifayati | *Next.js/React, Tailwind CSS, REST APIs, Vercel*

- Developed modular, accessible UI components with responsive layouts to ensure cross-device performance and sub-second load times.
- Architected a budget-conscious consumer platform focused on smart deal aggregation, price tracking, and cost optimization.
- Implemented client-side state management and asynchronous data fetching pipelines for seamless product and deal discovery.

FlowSuite | *JavaScript, React, Node.js, Express, REST APIs*

- Built an end-to-end workflow management suite designed to automate task lifecycles and streamline developer operational pipelines.
- Engineered secure RESTful endpoints for dynamic data ingestion, structured validation, and asynchronous task execution.
- Crafted a modular dashboard interface featuring responsive state management and real-time activity tracking.

PUBLICATIONS

Real-Time Intelligent Traffic Signal Control Using YOLOv8 with Ambulance Priority | *SSRN Electronic Journal*

- Published research proposing an AI-driven adaptive traffic signal framework integrating computer vision (YOLOv8) and dynamic queuing theory for prioritized emergency vehicle clearance.

CERTIFICATIONS & ACHIEVEMENTS

Anthropic Education: AI Fluency: Framework & Foundations (Sep 2026)

AWS Academy: Machine Learning Foundations (Feb 2024)

Oracle: Global Certification (Mar 2025)

Hackathons: Participant / Finalist at SRM Hackathon 8.0 (2024) and Layer 0 Hackathon (2023)